Regional and Urban Concentration Forces

LSE GY457 - AT 2025
Topic 1
Sara Bagagli

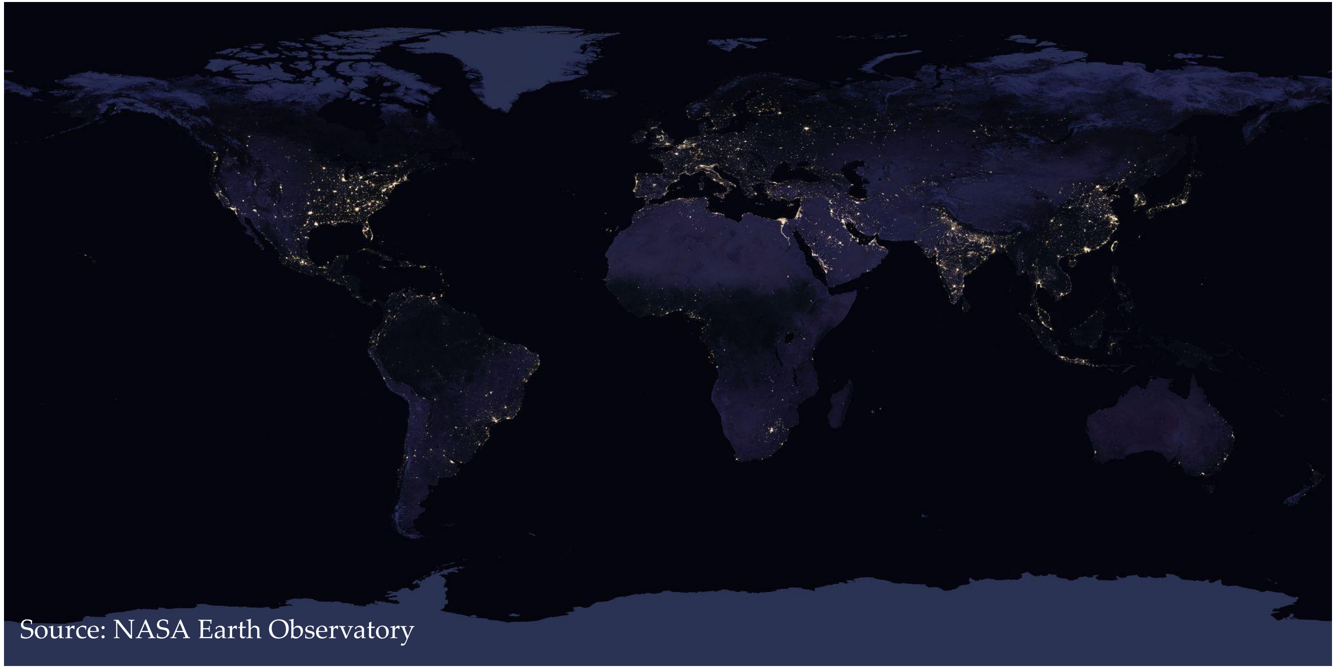
Logistics: Next week's seminar (Oct-15)

- Seminar allocation is available on Moodle
- Get ready for next week's seminar
- Reading: DiPasquale and Wheaton, Ch 1 (on Moodle)

Course overview

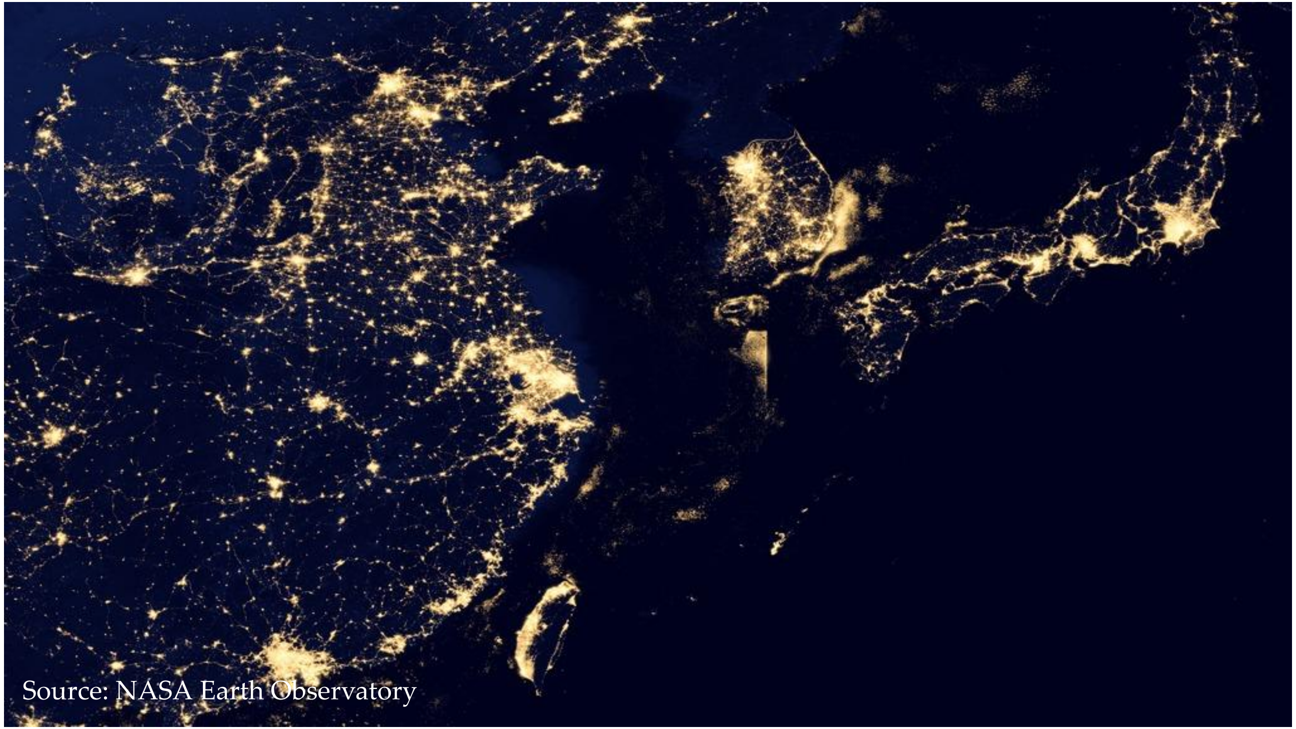
- Block 1
  - Intro to Urban and Regional Economics and course overview
  - **Topic 1: Regional and urban concentration forces**
  - Topic 2: The empirics of agglomeration
  - Topic 3: The costs and benefits of agglomeration
- Block 2
  - Topic 4&5: Monocentric city model
  - Topic 6: Firm location choice
- Block 3
  - Topic 7: The vertical dimension of cities
  - Topic 8: Neighborhoods and segregation
  - Topic 9: Gentrification and suburbanization
  - Topic 10: Measuring cities with new technologies

More than 50% of world's population lives on 1% of its land



Source: NASA Earth Observatory

Unequal spatial concentration



Unequal spatial concentration

Source: NASA Earth Observatory

Introduction

- Concentration of economic activity largely unequal
- Regional vs. urban concentration
- What drives the concentration of economic activity into regions?
- Why is it relevant to real estate markets?

Two questions

1. Why do large industrial regions exist?
   - Location fundamentals
   - Transportation costs (Krugman 1991)

2. Why do cities exist?
   - Location fundamentals
   - Agglomeration theory (Marshall)

Two questions

1. Why do large industrial regions exist?
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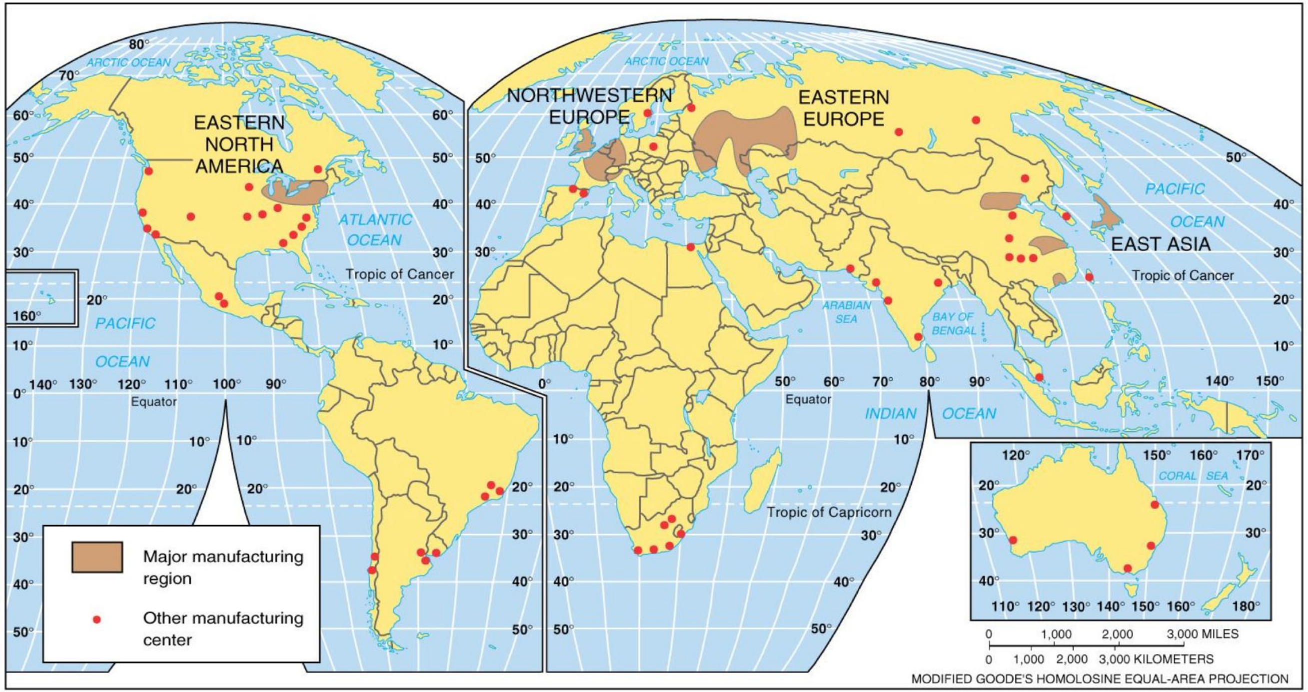
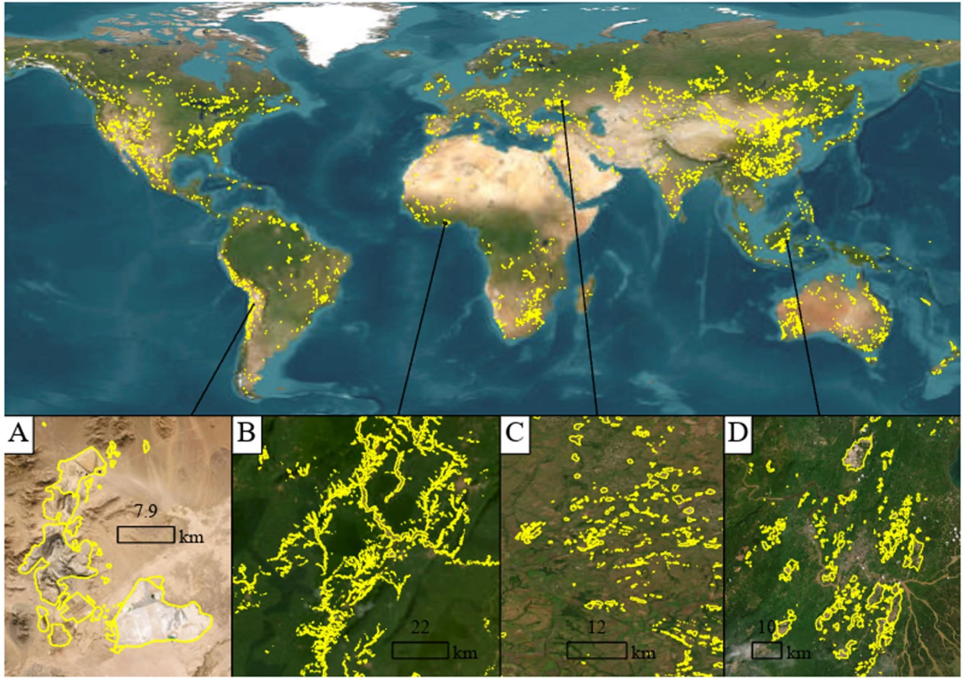
Why do large industrial regions exist

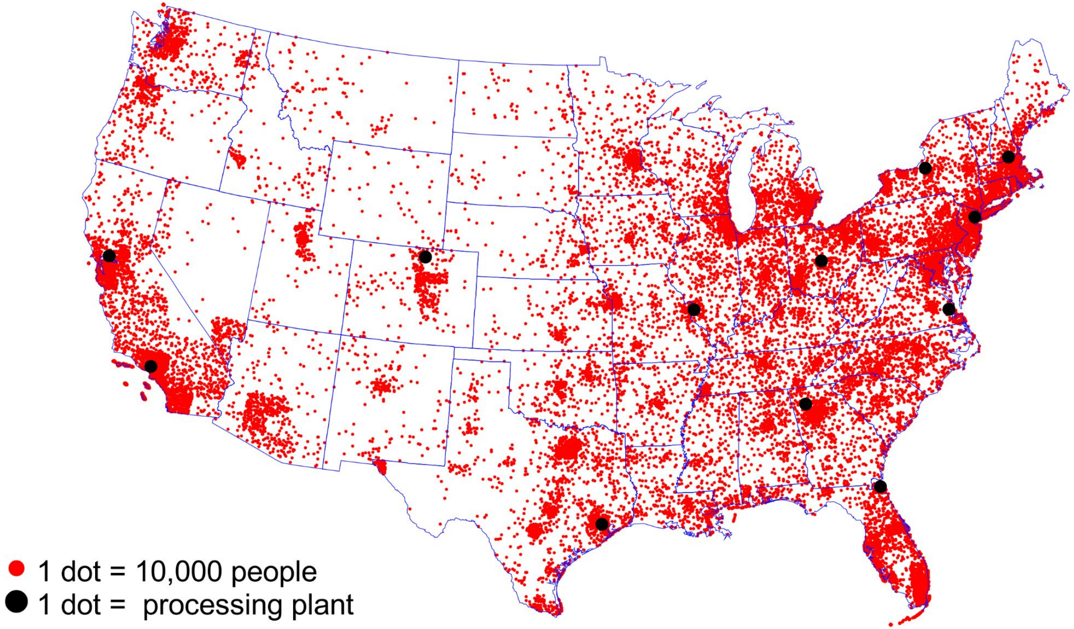


Global mine areas (Tang and Werner, 2023)

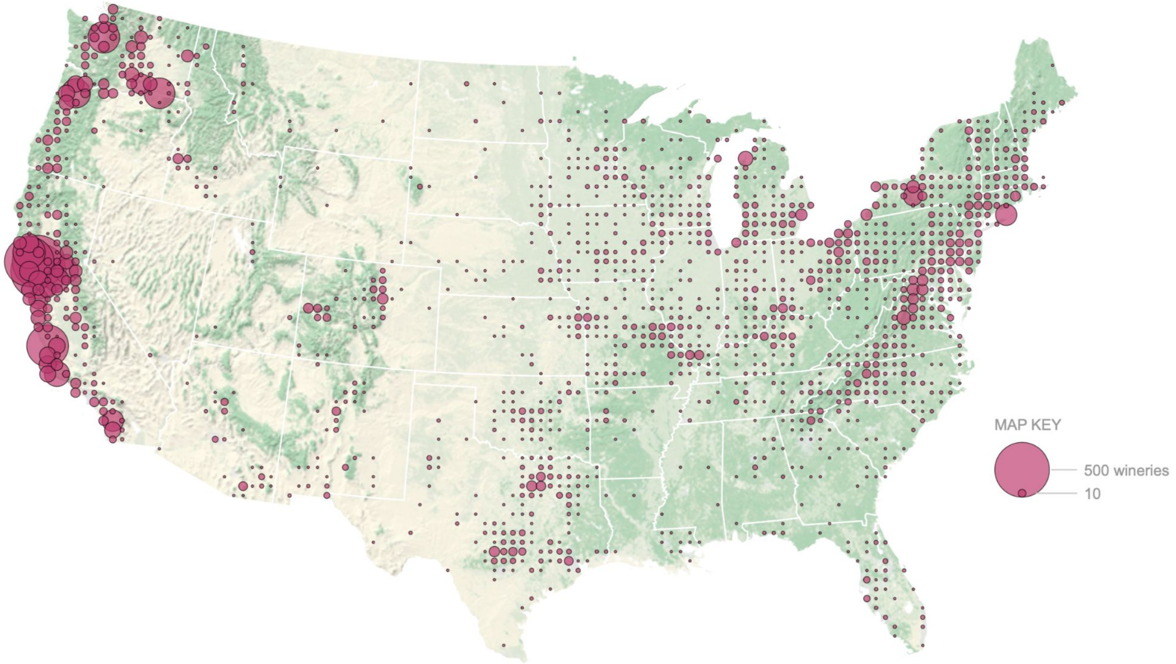


(A) Escondida, Chile (copper); (B) Ankobra River, Ghana (gold); (C) Donetsk, Ukraine (coal); (D) Samarinda, Indonesia (coal)

Location of Anheuser-Busch Breweries and Population 2000 (Holmes and Stevens, 2004)



Location of wineries



Source: “A nation of wineries” (New York Times, July 5, 2013)

Location fundamentals

- Some locations have high “fundamental value”
- Industrial regions often close to natural resources, like coal
- Emergence of such clusters often dates back to the Industrial Revolution (1760-1860)
    - Coal critical input to industrial production
- Access to natural resources historically important

Coal and the European Industrial Revolution (Fernihough and O'Rourke, 2021)

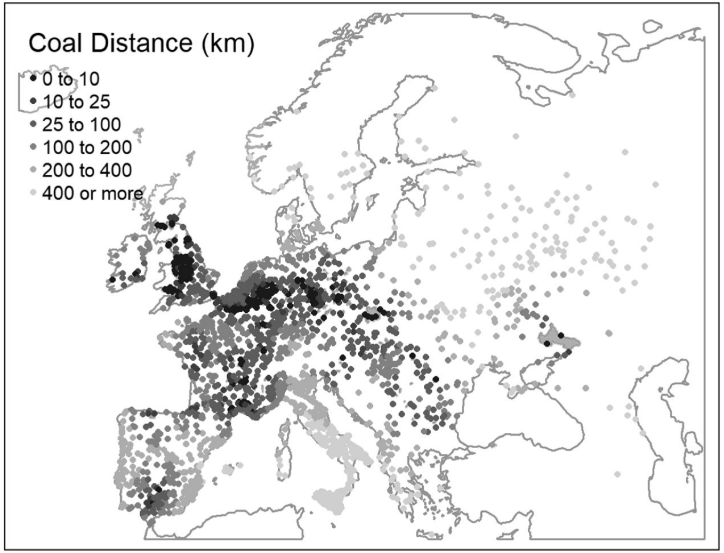


Figure: Cities' Distance to Coalfields

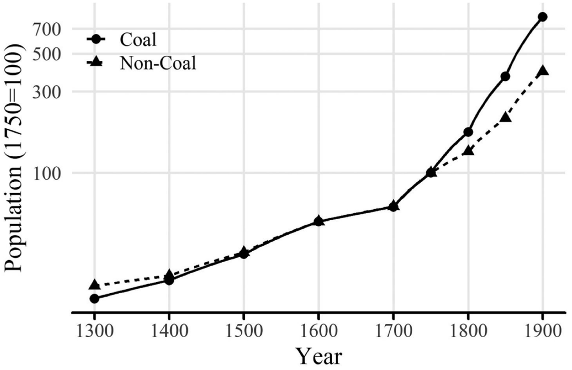


Figure: Population in cities located more or less than 25 km from Coalfields, 1300-1900

Fernihough and O’Rourke (2021)

- Before 1750, no relationship between proximity to coalfields and growth
- After 1750, cities close to coalfields grew faster than those far away
    - Cities within 49 km from the closest coalfield grew 21% faster than those 85 km further away

Fernihough and O’Rourke (2021)

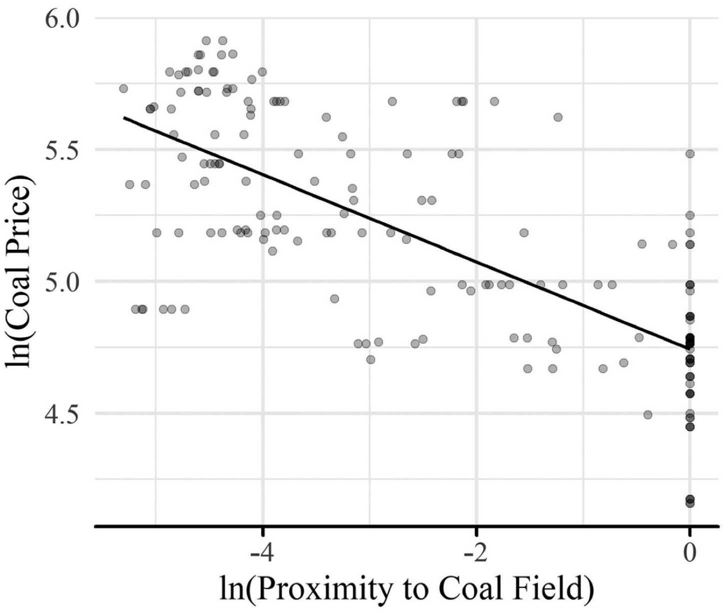


Figure: Proximity to coalfields and coal prices, UK 1843-1845

“Coal was bulky, heavy and costly to transport. It was also a fuel, whose weight vanished when it was used in the production process: there was thus substantial cost savings if coal was used close to where it was mined.”

Falling transportation costs

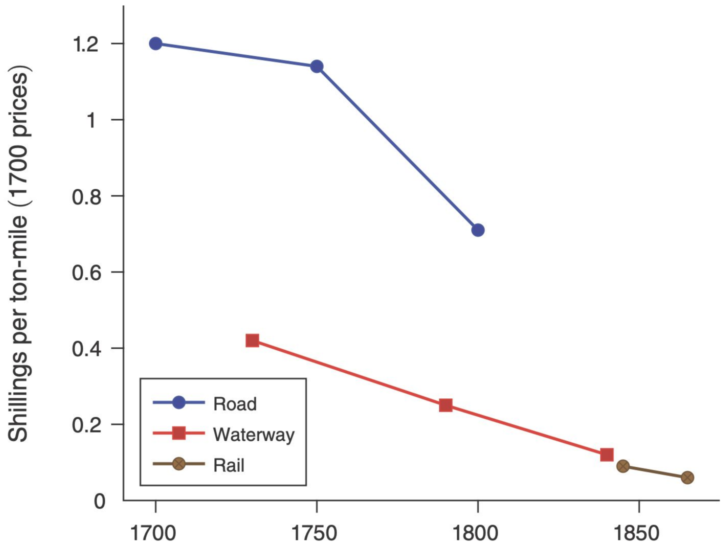


FIGURE 5. FREIGHT COSTS (BOGART 2014)

Source: Bogart (2014) and Trew (2020)

- Britain's "transport revolution"
- Reduction of 95% in the costs of transporting goods between 1700 and 1850 across all modes
- Naive estimates: Do not take into account shifts in mode along the network

Krugman (1991): Core-Periphery model

- Paul Krugman's seminal paper
- Key novelties: Locations are *dimension-full* + Increasing returns in production
- Ingredients:
  - Increasing returns in production
  - Transportation costs (trade costs)
  - Factor mobility
- What does each factor imply for the geography of an economy?

Krugman (1991): Core-Periphery model

- Increasing returns create an incentive to concentrate production
- (High) Transportation costs create an incentive to locate production close to consumers
- With increasing returns and low transportation costs, manufactures concentrate in a “core” from where they supply the agricultural “periphery”

Simple version of the core-periphery model

- Two regions, East (E) and West (W)
- Factors of production
  - 6 immobile farmers: 3 in each region
  - 4 mobile workers, in which region depends on the scenario
- Total population is 10
  - Each individual demands one unit of good inelastically
- Fixed cost for each manufacturing plant is 4
  - No variable costs
- Trade costs of manufactured good
  - 0 within region; 1 per unit of manufactured good across regions

Case A: Mobile workers in E & W

	Production in E		Production in E & W		Production in W	
	E	W	E	W	E	W
Farmers	3	3	3	3	3	3
Workers	2	2	2	2	2	2
Population	5	5	5	5	5	5
Goods shipped						
Fixed plant cost						
Shipping cost						
Total costs						

Where should production take place to minimize costs?

Case B: Mobile workers in E & W + Trade costs = 0.5

	Production in E		Production in E & W		Production in W	
	E	W	E	W	E	W
Farmers	3	3	3	3	3	3
Workers	2	2	2	2	2	2
Population	5	5	5	5	5	5
Goods shipped						
Fixed plant cost						
Shipping cost						
Total costs						

Where should production take place to minimize costs?

Summary of model's predictions

- Small shocks (e.g., reductions in transportation costs) can cause agglomeration in either region
- With high transportation costs, dispersion is the only outcome
- With low transportation costs, agglomeration
- For intermediate transportation costs, both outcomes are possible
- A priori, agglomeration can take place in any region, leaving a strong role to chance, history, expectations

Summary of model's predictions

- Krugman's Core-Periphery model a lot richer: (i) manufacturing workers move to regions with high wages; (ii) competition among firms affects prices and wages; (iii) market access affects firms' ability to pay high nominal wages
- But main insights of the interplay between transportation costs and agglomeration remain
- Framework allows to rationalize spatial concentrations across regions today

Two questions

1. Why do large industrial regions exist?
   - Location fundamentals
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2. Why do cities exist?
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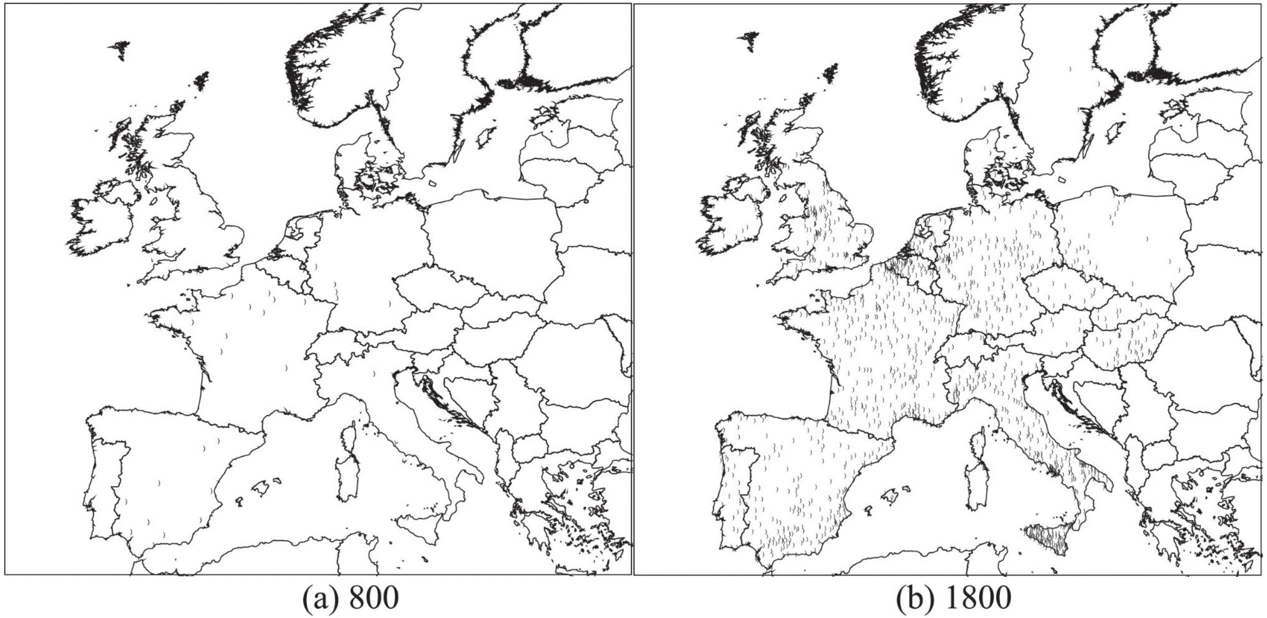
Trading cities

- A1: Comparative advantage (factor productivity)
  - Differences in geography, institutions, traditions,...
  - Locations (whether cities, regions, countries) specialize and trade
- A2: Scale economies in exchange
  - Indivisible inputs
  - Workers' specialization
- Favorable geography (like natural harbor)
- Concentration of trade workers (higher wages)
- Dominating concentration force before the industrial revolution (Athens, New York,...)

Bosker and Buringh (2017)



European city system in 800 and 1800

Bosker and Buringh (2017)

$P(city\ t \mid no\ city\ t-1)$	BASELINE
sea	0.007***
	[0.00]
river	0.012***
	[0.00]
hub	0.069***
	[0.00]
road	0.004***
	[0.00]
ln elevation	+0.00
	[0.98]
ruggedness	0.0001***
	[0.00]
P(cultivation)	0.0002**
	[0.02]
city >= 10k? $(t-1)$	
0–20 km	−0.0005***
	[0.00]
20–50 km	0.0002**
	[0.02]
50–100 km	0.0002***
	[0.00]

Determinants of city formation in 800-1800 Europe:

1. Being close to sea/river; Roman road/road hub
2. Being between 20-100 km of a big city
   - Good: Trade
3. BUT: Negative being within 20 km of a big city
   - Bad: Spatial competition

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1 = First nature geography
2,3 = Second nature geography

Factory cities

- A1: Economies of scale in production
  - Indivisible inputs, machines, fixed costs
  - Worker specialization
- Higher productivity
  - Higher wages, higher rents
  - More workers within commuting range
- Factory towns
  - Trade industrial with agricultural products
- Dominating concentration force during industrialization (as we have seen)

Agglomeration economies

- Alfred Marshall (1890) "Principle of Economics"
- There exist city-specific costs
  - Congestion, higher rents
  - Higher wages (required to compensate for rents in equilibrium)
- Firms in cities must be more productive to compensate for these costs
  - Emphasis on returns *external* to firms
  - As opposed to *internal returns* as in Krugman 1991
- Cities benefit from co-location of goods, people, ideas
- Transportation of people over short distances expensive because of value time

Localization vs. urbanization economies

- 2 types of place-specific returns to scale external to firm
- Localization economies, related to specialization within industries in clusters
  - e.g., autos in Detroit, finance in London
- Urbanization economies (Jacobs, 1969), related to interaction across (complementary) services – diversity
  - Various services related to production activity, like legal, real estate, leisure,...
- Different from place-specific internal returns, due to investments into single large factories

Agglomeration economies: Marshallian externalities

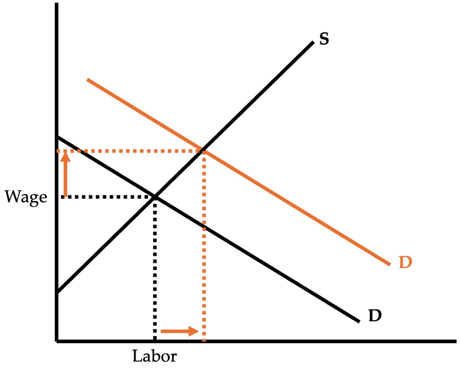
- Shared inputs (proximity of goods)
  - Cheaper intermediate inputs and services (suppliers close by)
- Labor market pooling (proximity of people)
  - Sorting (more productive workers in cities)
  - Matching (better firms-workers match)
  - Learning (workers learn from each other)
- Knowledge spillovers (proximity of ideas)
  - Access to non-codified knowledge
  - Leads to innovation and productivity

Shared inputs

- A1: Conventional inputs
  - Raw material, labor, capital (reasonably mobile)
- A2: Non-traded local (intermediate) inputs
  - Specialized, locally available inputs (high Weber transport costs)
  - e.g., producers of specialist legal services,...
  - Indivisible inputs (high fixed costs relative to marginal costs)
  - Access only via location in the city
- Services offered at lower cost in cluster
- Cost accessing intermediate inputs lower in clusters

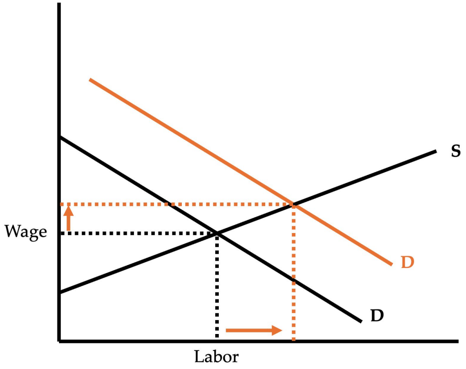
Labor market pooling: Sorting

- Labor acquisition cost: Hire qualified workers relatively quickly and cheaply in “thick markets” (elastic supply)



SMALL CLUSTER

Increase in labor demand > small increase in quantity (labor) and large increase in price (wage)



LARGE CLUSTER

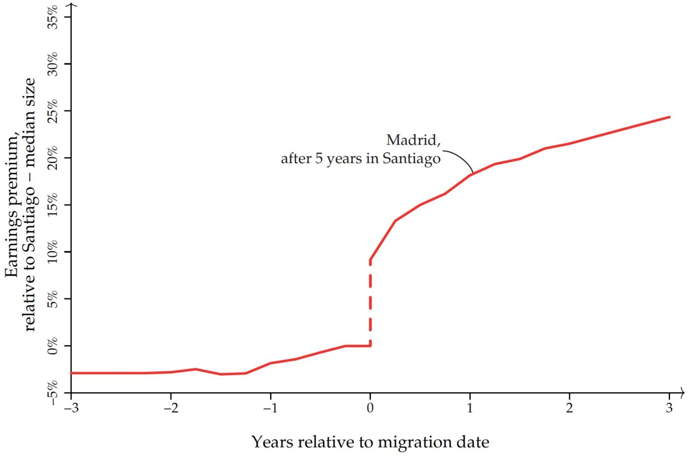
Increase in labor demand > large increase in quantity (labor) and small increase in price (wage)

Labor market pooling: Matching

- Matching
  - Improved quality of the match
- Positive assortative matching
  - More productive firms match to more productive workers
  - More efficient matches in bigger cities
  - E.g., Dauth et al. (2022) use matched employer-employee data for Germany and show that “high-quality” workers are more likely to work for “high-quality” firms in denser cities

Labor market pooling: Learning

- Learning in larger cities
    - Workers learn from one another a variety of skills and experiences



- Learn faster in bigger cities (De La Roca and Puga, 2017)
- Figure compares earnings of worker in Santiago that then moves to Madrid with earnings of worker that stays in Santiago

Knowledge spillovers

- Interactions among different firms
  - Jacobs (1969)
- Transmission via face-to-face contacts and probabilistic encounters
  - e.g., Davis and Dingle (2019)
- Information shared on non-market basis
  - e.g., products, production processes, technology,...
- Access via location in specialized clusters (“City”, “Silicon Valley”)

Why do cities exist? (Duranton and Puga, 2004)

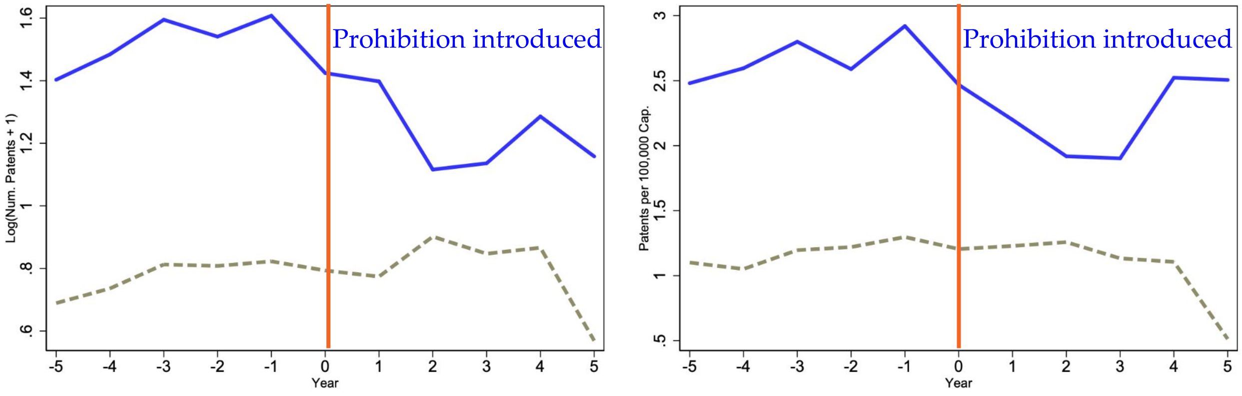
- Why do firms and workers cluster in cities?
- Increasing returns essential to understand why there are cities
- Cities are the trade-off between agglomeration economies (or localized aggregate increasing returns) and the costs of urban congestion
- Identify micro-foundations of urban agglomeration economies

Duranton and Puga (2004)

- Three types of micro-foundations
  - Based on sharing, matching, and learning mechanisms
- Sharing indivisible facilities, risks, input suppliers,...
- Matching: agglomeration improves quality/likelihood of a match (e.g., between workers and firms)
- Learning deals with generation, diffusion, accumulation of knowledge

Andrews (2020)

- Bar talks, alcohol prohibition, and invention



Number patents (log) Patents per capita

- Idea: Prohibition disrupts informal social interactions that foster innovation

Summary

- Incentives for spatial clustering in regions
  - Increasing returns + low transportation costs (for goods)
- Incentives for spatial clustering in cities
  - External returns + high transportation costs (for people)
- Impact on real estate
  - Clustering increases the demand for real estate, shifting the demand curve
  - Changes in productivity origins/sources affect future demand

Next week

- The empirics of agglomeration
  - Measurement
  - Clustering and the determinants of agglomeration
  - Spatial scale of agglomeration effects
- The effects of agglomeration
- Urbanization vs. localization effects
  - What matters for urban growth

Thank you!

Readings and references

Required readings

- O'Sullivan, Ch. 1-3
- Krugman, P. (1991), Geography and Trade, Ch. 1, Center and Periphery.

Other readings and references

- Andrews, M. (2020). Bar Talks: Informal Social Interactions, Alcohol Prohibition, and Invention. SSRN.
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Readings and references

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Appendix

Case C: All mobile workers in E

	Production in E		Production in E & W		Production in W	
	E	W	E	W	E	W
Farmers	3	3	3	3	3	3
Workers	4	0	4	0	4	0
Population	7	3	7	3	7	3
Goods shipped	3		0		7	
Fixed plant cost	4		8		4	
Shipping cost	3		0		7	
Total costs	7		8		11	

Cost-minimizing option is concentration of production in E: Agglomeration

Case D: All mobile workers in E + Trade costs = 0.5

	Production in E		Production in E & W		Production in W	
	E	W	E	W	E	W
Farmers	3	3	3	3	3	3
Workers	4	0	4	0	4	0
Population	7	3	7	3	7	3
Goods shipped	3		0		7	
Fixed plant cost	4		8		4	
Shipping cost	1.5		0		3.5	
Total costs	5.5		8		7.5	

Cost-minimizing option is concentration of production in E: Agglomeration

Case E: All mobile workers in E + Trade costs = 2

	Production in E		Production in E & W		Production in W	
	E	W	E	W	E	W
Farmers	3	3	3	3	3	3
Workers	4	0	4	0	4	0
Population	7	3	7	3	7	3
Goods shipped	3		0		7	
Fixed plant cost	4		8		4	
Shipping cost	6		0		14	
Total costs	10		8		18	

Cost-minimizing option is concentration of production in E & W: Dispersion

Case F: Mobile workers in E & W + Trade costs = 2

	Production in E		Production in E & W		Production in W	
	E	W	E	W	E	W
Farmers	3	3	3	3	3	3
Workers	2	2	2	2	2	2
Population	5	5	5	5	5	5
Goods shipped	5		0		5	
Fixed plant cost	4		8		4	
Shipping cost	10		0		10	
Total costs	14		8		14	

Cost-minimizing option is concentration of production in E & W: Dispersion